

Katherine Sandeen

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Education

Oregon State University, BS Computer Science

Expected Dec 2025

GPA: 4.0 | Software Engineering, Data Structures, Databases, Web Development, Algorithms

University of North Florida, BFA

Apr 2015

Experience

Software Engineering Intern | Tokyo Electron Company

May 2025 – August 2025

- Developed a web-based backup for cleanroom and Metrology job tracking to mitigate downtime in a 24-hour semiconductor R&D facility, ensuring access to job data within 5 minutes of system blackout.
- Implemented a new backend in C#/.NET with GraphQL, including database migration, schema design, and queries to support read operations.
- Built a responsive interface using React, TypeScript, and layout mockups, applying component-based architecture to streamline UX and prepare the foundation for full system replacement.

Teaching Assistant: Data Structures | Oregon State University

Apr 2024 – September 2025

- Collaborated with faculty to optimize learning for a cohort of 300+ students, boosting overall course performance.
- Mentored students during 6 large data structures projects dealing with complex concepts such as AVL trees and hash maps.
- Facilitated discussions and provided technical support during biweekly office hours, delivering one-on-one support to 60+ students

Equipment Technician & Consultant | Georgie's Ceramic & Clay Co.

Oct 2016 – May 2025

- Analyzed data to troubleshoot equipment issues, accomplishing a 95% resolution rate and increasing client satisfaction.
- Led monthly technical workshops, attracting an average of 100+ participants each year, resulting in increased community engagement and client confidence.
- Partnered with a diverse team of 5+ specialists to elevate client experiences, using user-centric problem-solving strategies.

Technical Skills

Languages: TypeScript, JavaScript, HTML, CSS, SQL, C#/.NET, Python

Libraries/Frameworks: React, Node.js, Express.js, Next.js, Chakra UI, Material UI

Tools/Platforms: Git, GitHub, Figma, Postman

Databases: GraphQL, SSMS, MongoDB

Projects

Website: CareLink

github.com/SandKat214/CareLink

JavaScript, React, Express.js, Node.js, MongoDB, Chakra UI, Figma, Cloudinary

- Collaborated with a 3-person team to build and optimize 1 microservices, following agile practices to ensure efficient development.
- Developed a secure patient management and scheduling app using React for a responsive user interface, and MongoDB for storing patient data, appointments, and history.
- Coded 4 microservices (login authentication, calendar management, email notifications, and image storage) in Node.js, improving modularity and maintainability.

Website: Web Dev Portfolio

github.com/SandKat214/WebDevPortfolio

JavaScript, HTML, CSS, React, Express.js, Node.js, MongoDB

- Created a website for art collection management using React and CSS for front-end styling, cutting loading times and enhancing user experience.
- Employed a RESTful API, boosting data handling efficiency and refining user accessibility.
- Utilized Node.js and MongoDB for efficient backend operations and data storage.